circumference must appear completely dark, when viewed from far away; it must be what we now call a black hole. The bottom line is the same, but the mechanism is completely different:

Michell and Laplace, with their Newtonian view of space and time as absolute and the speed of light as relative, believed that for a star just a bit smaller than the critical circumference, corpuscles of light would very nearly escape. They would fly up to great heights above the star, higher than any orbiting planet; but as they climbed, they would be slowed by the star's gravity, then halted somewhere short of interstellar space, then turned around and pulled back down to the star. Though creatures on an orbiting planet could see the star by its slow-moving light (to them it would not be dark), we, living far away on Earth, could not see it at all. The star's light could not reach us. For us the star would be totally black.

By contrast, Schwarzschild's spacetime curvature required that light always propagate with the same universal speed; it can never be slowed. (The speed of light is absolute, but space and time are relative.) However, if emitted from the critical circumference, the light must get shifted in wavelength an infinite amount, while traveling upward an infinitesimal distance. (The wavelength shift must be infinite because the flow of time is infinitely dilated at the horizon, and the wavelength always shifts by the same amount as time is dilated.) This infinite shift of wavelength, in effect, removes all the light's energy; and the light, thereupon, ceases to exist! Thus, no matter how close a planet might be to the critical circumference, creatures on it cannot see any light at all emerging from the star.

In Chapter 7, we shall study how the light behaves as seen from inside a black hole's critical circumference, and shall discover that it does not cease to exist after all. Rather, it simply is unable to escape the critical circumference (the hole's horizon) even though it is moving outward at the standard, universal speed of 299,792 kilometers per second. But this early in the book, we are not yet ready to comprehend such seemingly contradictory behavior. We must first build up our understanding of other things, as did physicists during the decades between 1916 and 1960.

During the 1920s and into the 1930s, the world's most renowned experts on general relativity were Albert Einstein and the British astrophysicist Arthur Eddington. Others understood relativity, but Einstein and Eddington